Bitcoin X-Sentiment Price Prediction — Evaluation Report

Out-of-sample evaluation window: **2026-07-10** → **2026-11-03** (UTC daily bars). Primary model chosen before testing: **Ensemble (primary)**. Transaction cost 10 bp per unit of turnover; risk-free rate 0; annualisation √365.

1. Headline results

- **Trading the n+0 forecast (long/flat):** Sharpe **-1.40**, Sortino **-2.17**, Max drawdown **-17.3%**, total return -10.8% with 40.2% time in market.
- **Buy & hold over the same days:** Sharpe 1.88, Sortino 3.47, Max drawdown -11.5%, total return +22.4%.
- **Forecast skill vs random walk:** RMSE ratio ranges 1.041–1.115 across n+0…n+5 (<1 beats the random walk); directional accuracy 40.2%–45.0%.
- **Statistical significance:** the Diebold–Mariano test rejects 'no better than random walk' at 5% for 0 of 6 horizons; it finds the model significantly *worse* than the random walk at 4 (n+0, n+1, n+2, n+3).
- **Does X sentiment add information?** Adding sentiment features to the same Ridge model lowered out-of-sample error at 1 of 6 horizons (ablation vs price-only).

2. Data

Source	Posts	Flagged spam	Days	First	Last
`apify:kaitoeasyapi/twitter-x-data-tweet-scraper-pay-per-result-cheapest`	9,842	344	181	2026-05-08	2026-11-04

Prices: 240 daily bars (2026-03-09 → 2026-11-03), source `yfinance`, ticker `BTC-USD`. News: 44 Tavily articles.

Apify runs:

- actor `kaitoeasyapi/twitter-x-data-tweet-scraper-pay-per-result-cheapest` — SUCCEEDED: 180 windows, 9,842 posts

X query: `(bitcoin OR BTC OR $BTC OR #bitcoin OR #BTC)` (ticker, cashtag and keyword in one advanced search), language en, 24-hour windows capped at 80 posts, sort `Top`. Posts are bucketed by UTC date to match the 00:00–24:00 UTC daily bar; an America/New_York timestamp is stored alongside for reference.

3. Method

Timing (no look-ahead). Row *n* is the current day; the forecast is made at its 00:00 UTC open using posts and prices up to the close of day *n−1* only. Targets are `log(close[n+h] / close[n−1])` for h = 0…5, so *n+0* is today's close and *n+5* the close five days later. Forecast prices are `close[n−1] · exp(ŷ)`.

Sentiment. VADER (built for social-media text) extended with a crypto lexicon (moon, rekt, HODL, FUD, dump, ETF inflows/outflows, …). URLs and @mentions are stripped; cashtag-stuffed shill posts (> 5 cashtags), same-day duplicates and near-empty posts, and giveaway / engagement-farming posts ("repost & reply with your BTC address") are flagged as spam and excluded.

Features — sentiment (13): engagement-weighted score and its rolling 30-day z-score, follower-weighted and influencer (≥10k followers) z-scores, bull−bear share, dispersion, 1-day change, 3/7-day EMA momentum, 7-day mean, post-volume and engagement z-scores, sentiment×engagement. **Price (13):** 1/2/3-day lagged returns, 5/10/20-day momentum, 7/20-day realised volatility, RSI-14, gap to the 20-day mean, high-low range, day-of-week. No traded volume, on-chain or macro inputs.

Models. One model per horizon: Ridge (regularisation chosen by time-ordered CV with a gap of h days) and depth-2 gradient-boosted trees on sentiment + price, their average (the *ensemble*, fixed in advance as primary), a price-only Ridge as an ablation, and random-walk / drift baselines.

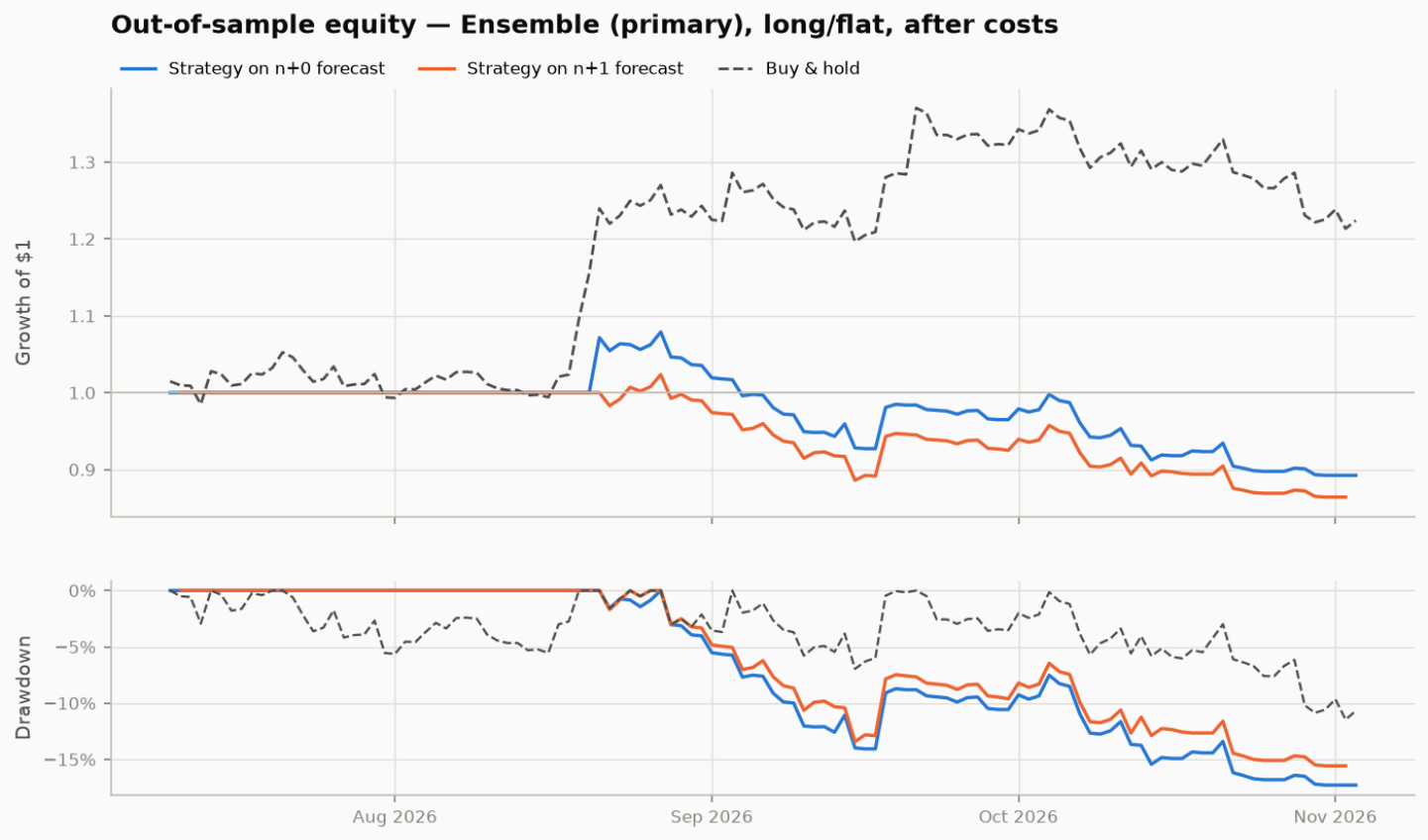
Walk-forward validation. Expanding window, minimum 60 training days, refit every 1 day(s). For horizon *h*, training rows are purged so every training target is already realised at the forecast time (row *i* is used only if $i + h \le n - 1$). Every prediction in this report is out of sample.

Backtest. At each open, go long if the forecast log-return > 0, otherwise flat; hold one day; pay costs on every change of position. Each horizon's forecast is traded as its own signal so positions never overlap.

4. Trading performance (out of sample, after costs)

Signal	Sharpe	Sortino	Max DD	Total return	CAGR	Ann. vol	Calmar	Exposure	Hit rate	Trades
Buy & hold	1.88	3.47	-11.5%	+22.4%	+87.8%	37.2%	7.66	100.0%	49.6%	1
Ensemble (primary) · n+0	-1.40	-2.17	-17.3%	-10.8%	-29.9%	23.4%	-1.73	40.2%	42.6%	36
Ridge (price only) · n+0	-1.62	-2.41	-12.0%	-9.2%	-25.9%	17.6%	-2.16	24.8%	37.9%	16
Ensemble (primary) · n+1	-2.22	-2.92	-15.6%	-13.6%	-37.1%	20.0%	-2.38	47.0%	42.6%	26
Ridge (price only) · n+1	-1.93	-2.26	-7.7%	-7.7%	-22.3%	12.7%	-2.92	23.5%	44.4%	4
Ensemble (primary) · n+2	-0.20	-0.32	-12.3%	-2.8%	-8.6%	27.1%	-0.70	56.6%	45.3%	8
Ridge (price only) · n+2	-1.97	-2.30	-7.3%	-7.3%	-21.8%	12.1%	-2.97	22.1%	40.0%	10
Ensemble (primary) · n+3	-0.05	-0.08	-12.2%	-1.5%	-4.9%	27.3%	-0.41	56.8%	46.0%	10

Signal	Sharpe	Sortino	Max DD	Total return	CAGR	Ann. vol	Calmar	Exposure	Hit rate	Trades
Ridge (price only) · n+3	-1.92	-2.28	-7.8%	-7.8%	-23.4%	13.4%	-3.01	30.6%	41.2%	2
Ensemble (primary) · n+4	-0.68	-0.97	-10.9%	-5.6%	-17.6%	24.3%	-1.61	54.1%	47.5%	7
Ridge (price only) · n+4	-1.94	-2.31	-7.8%	-7.8%	-23.8%	13.5%	-3.05	32.1%	40.0%	2
Ensemble (primary) · n+5	-0.75	-1.07	-10.3%	-6.1%	-19.2%	24.4%	-1.87	54.2%	48.3%	5
Ridge (price only) · n+5	-2.14	-2.44	-10.1%	-9.8%	-29.8%	15.9%	-2.96	34.6%	43.2%	1

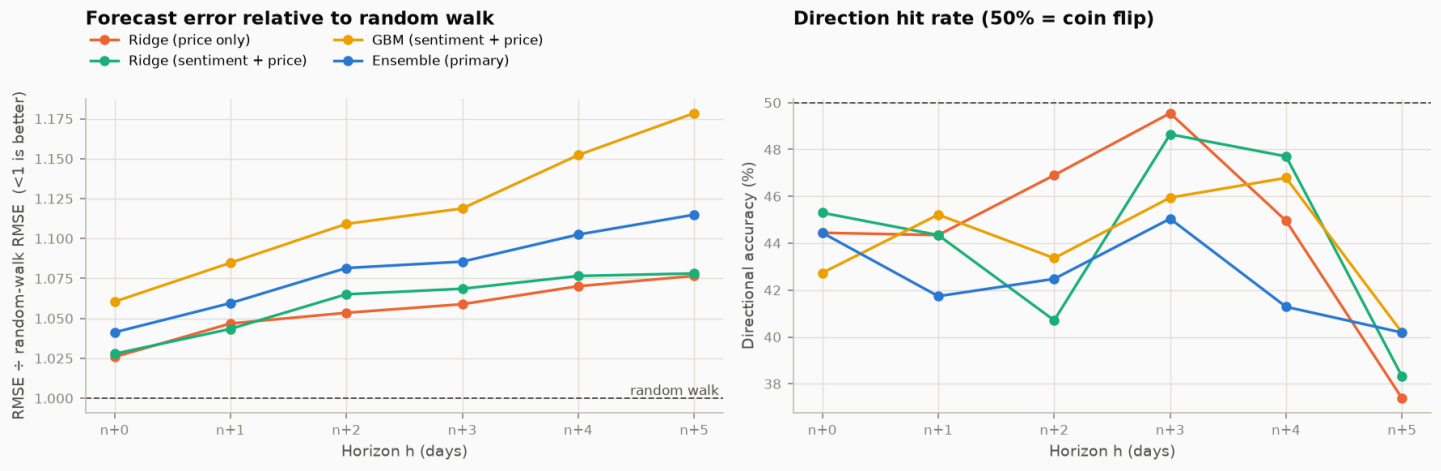


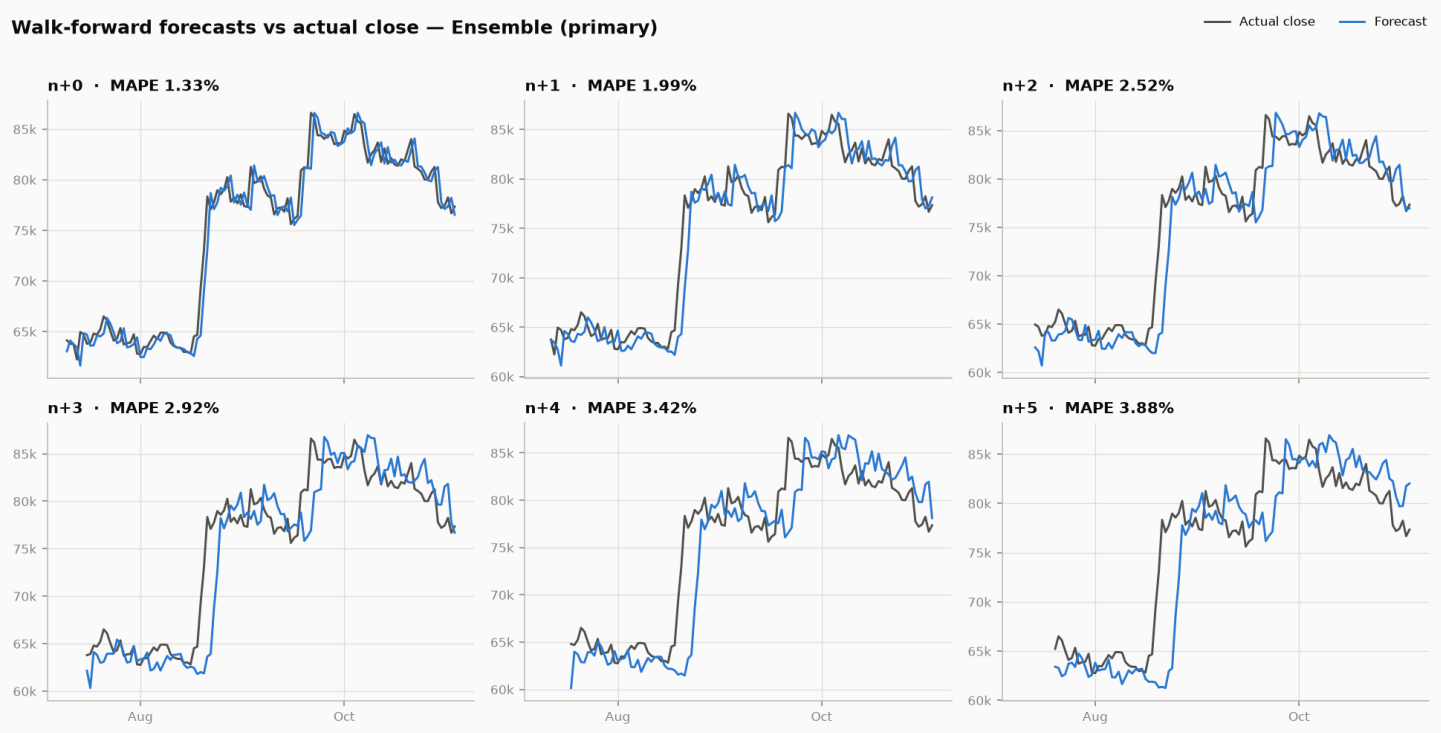
5. Forecast accuracy by horizon

Horizon	Model	N	MAPE %	RMSE $	RMSE ÷ RW	Dir. acc.	p (dir)	IC	DM stat	DM p
n+0	Ensemble (primary)	117	1.33	$1,503	1.04	44.4%	0.90	-0.24	2.84	0.00
n+0	GBM (sentiment + price)	117	1.38	$1,530	1.06	42.7%	0.95	-0.27	3.24	0.00
n+0	Ridge (sentiment + price)	117	1.31	$1,484	1.03	45.3%	0.87	-0.18	2.13	0.03
n+0	Ridge (price only)	117	1.30	$1,479	1.03	44.4%	0.90	-0.14	1.82	0.07
n+0	Random walk	117	1.26	$1,450	1.00	–	–	–	–	–
n+1	Ensemble (primary)	115	1.99	$2,168	1.06	41.7%	0.97	-0.18	2.63	0.01
n+1	GBM (sentiment + price)	115	2.08	$2,217	1.08	45.2%	0.87	-0.19	3.00	0.00
n+1	Ridge (sentiment + price)	115	1.94	$2,138	1.04	44.3%	0.90	-0.19	2.40	0.02
n+1	Ridge (price only)	115	1.94	$2,139	1.05	44.3%	0.90	-0.18	2.13	0.03
n+1	Random walk	115	1.86	$2,061	1.00	–	–	–	–	–
n+2	Ensemble (primary)	113	2.52	$2,779	1.08	42.5%	0.95	-0.24	2.58	0.01
n+2	GBM (sentiment + price)	113	2.63	$2,849	1.11	43.4%	0.93	-0.21	2.79	0.01
n+2	Ridge (sentiment + price)	113	2.43	$2,740	1.07	40.7%	0.98	-0.30	2.52	0.01
n+2	Ridge (price only)	113	2.36	$2,705	1.05	46.9%	0.77	-0.24	2.12	0.03
n+2	Random walk	113	2.27	$2,590	1.00	–	–	–	–	–
n+3	Ensemble (primary)	111	2.92	$3,361	1.09	45.0%	0.87	-0.16	2.02	0.04
n+3	GBM (sentiment + price)	111	3.13	$3,463	1.12	45.9%	0.83	-0.12	2.20	0.03
n+3	Ridge (sentiment + price)	111	2.78	$3,315	1.07	48.6%	0.65	-0.28	2.14	0.03
n+3	Ridge (price only)	111	2.73	$3,280	1.06	49.5%	0.58	-0.27	1.84	0.07
n+3	Random walk	111	2.63	$3,125	1.00	–	–	–	–	–
n+4	Ensemble (primary)	109	3.42	$3,952	1.10	41.3%	0.97	-0.13	1.77	0.08

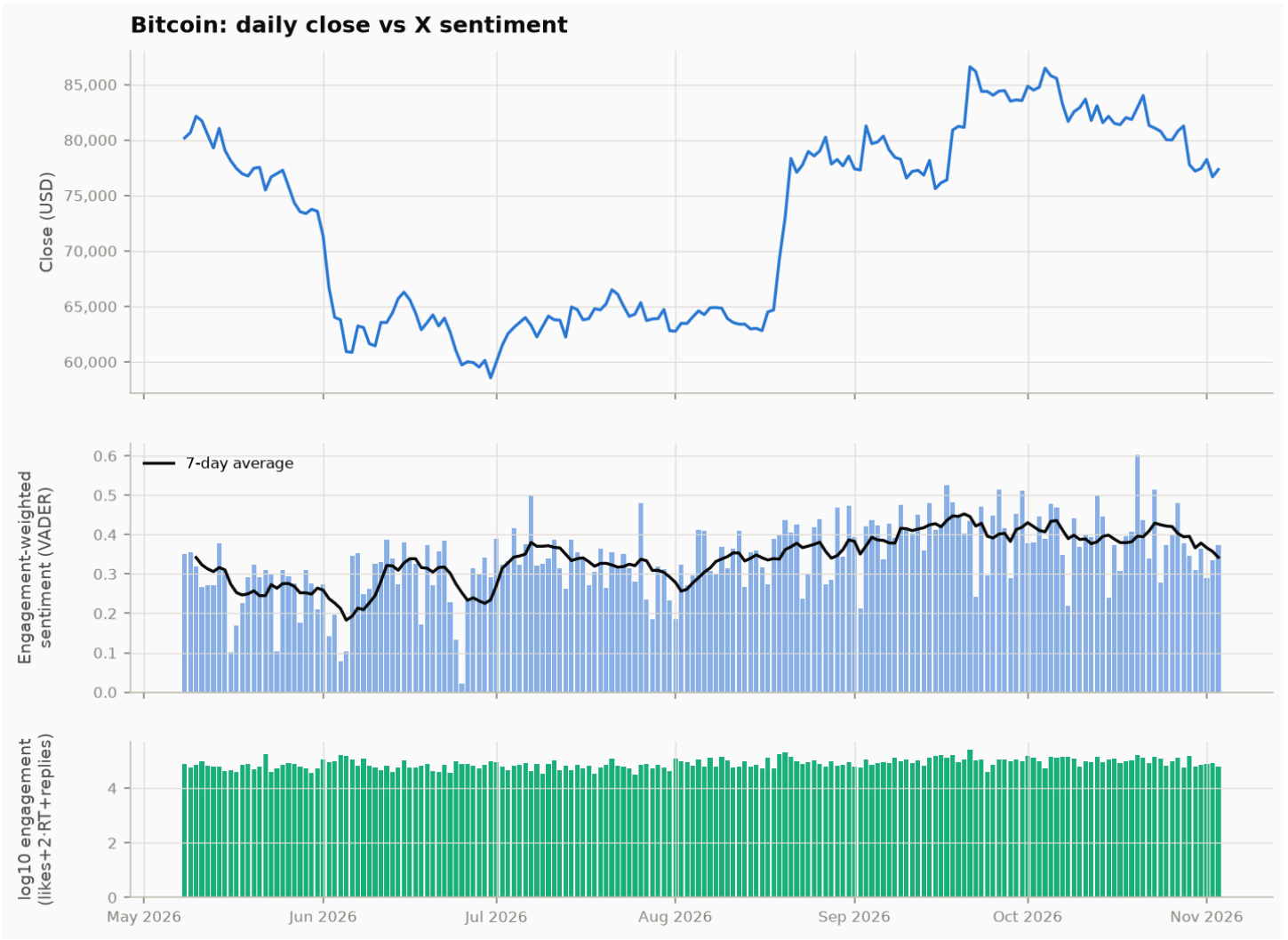
Horizon	Model	N	MAPE %	RMSE $	RMSE ÷ RW	Dir. acc.	p (dir)	IC	DM stat	DM p
n+4	GBM (sentiment + price)	109	3.72	$4,131	1.15	46.8%	0.78	-0.10	1.95	0.05
n+4	Ridge (sentiment + price)	109	3.23	$3,869	1.08	47.7%	0.72	-0.31	1.95	0.05
n+4	Ridge (price only)	109	3.18	$3,841	1.07	45.0%	0.87	-0.28	1.78	0.08
n+4	Random walk	109	3.03	$3,626	1.00	–	–	–	–	–
n+5	Ensemble (primary)	107	3.88	$4,477	1.11	40.2%	0.98	-0.13	1.63	0.10
n+5	GBM (sentiment + price)	107	4.24	$4,742	1.18	40.2%	0.98	-0.12	1.85	0.06
n+5	Ridge (sentiment + price)	107	3.65	$4,336	1.08	38.3%	0.99	-0.28	1.71	0.09
n+5	Ridge (price only)	107	3.65	$4,328	1.08	37.4%	1.00	-0.27	1.67	0.10
n+5	Random walk	107	3.40	$4,063	1.00	–	–	–	–	–

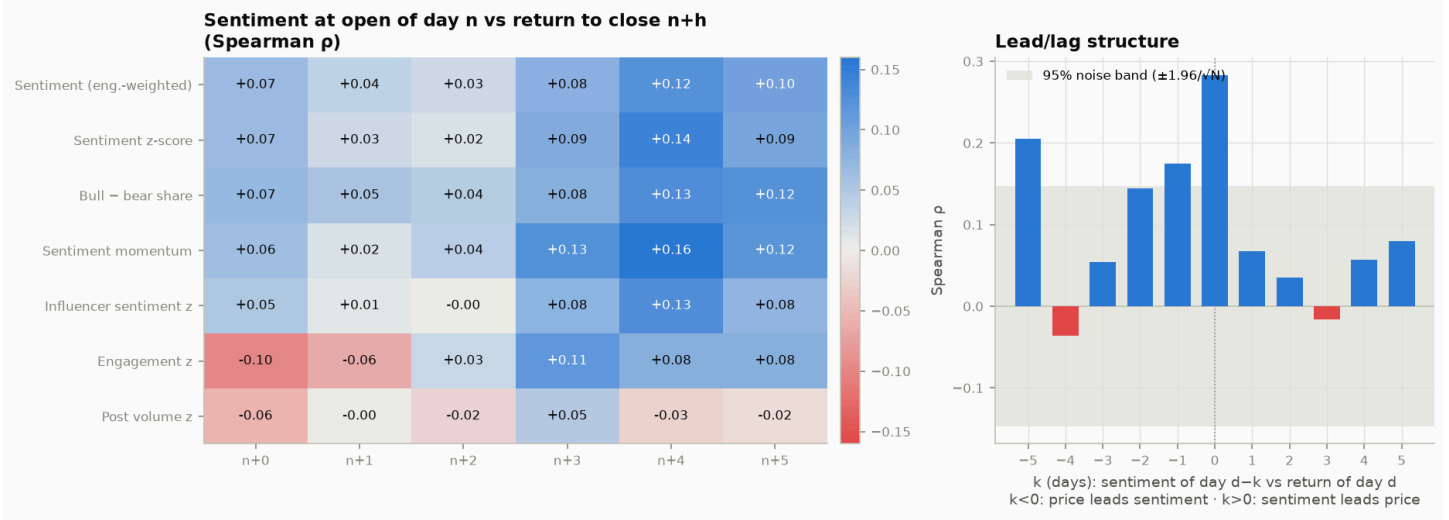
RMSE ÷ RW < 1 means lower error than the random walk. DM = Diebold–Mariano test on squared errors vs random walk (Newey–West, lag h); negative = model better. p (dir) = one-sided binomial test of the hit rate against 50%.

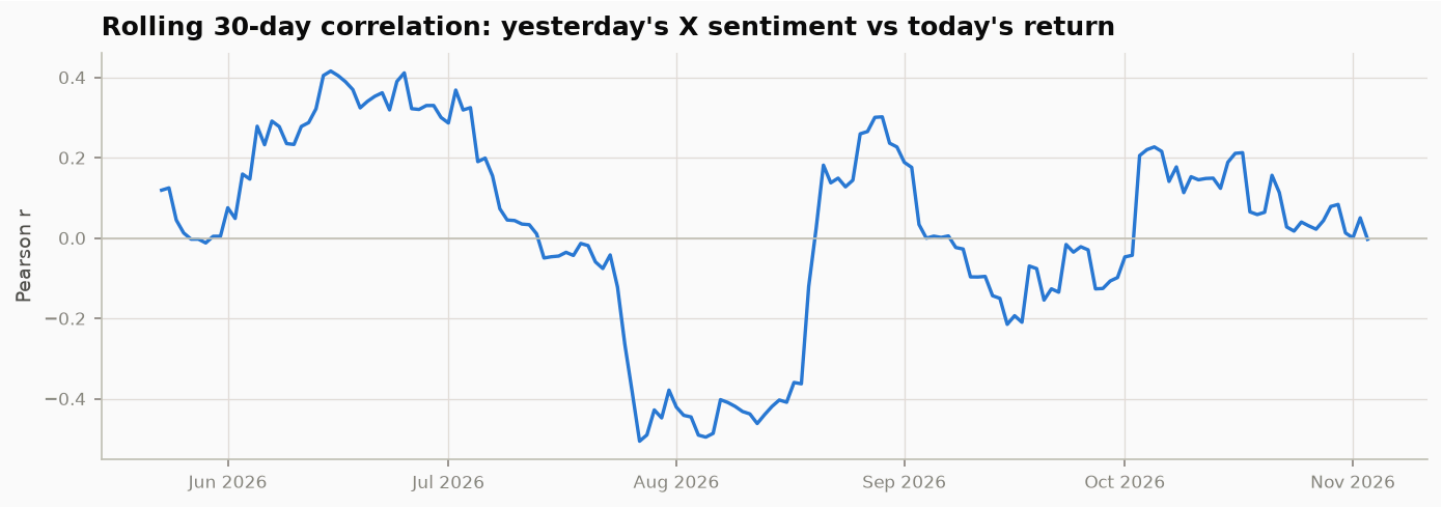




6. Sentiment vs price


Bitcoin: daily close vs X sentiment
Close (USD)
85,000
80,000
75,000
70,000
65,000
60,000
Engagement-weighted sentiment (VADER)
7-day average
0.6
0.5
0.4
0.3
0.2
0.1
0.0
log10 engagement (likes+2·RT+replies)
4
2
0
May 2026
Jun 2026
Jul 2026
Aug 2026
Sep 2026
Oct 2026
Nov 2026


Sentiment at open of day n vs return to close n+h
(Spearman ρ)
Sentiment (eng.-weighted)
Sentiment z-score
Bull − bear share
Sentiment momentum
Influencer sentiment z
Engagement z
Post volume z
+0.07 +0.04 +0.03 +0.08 +0.12 +0.10
+0.07 +0.03 +0.02 +0.09 +0.14 +0.09
+0.07 +0.05 +0.04 +0.08 +0.13 +0.12
+0.06 +0.02 +0.04 +0.13 +0.16 +0.12
+0.05 +0.01 -0.00 +0.08 +0.13 +0.08
-0.10 -0.06 +0.03 +0.11 +0.08 +0.08
-0.06 -0.00 -0.02 +0.05 -0.03 -0.02
n+0
n+1
n+2
n+3
n+4
n+5
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Lead/lag structure
95% noise band (±1.96/√N)
Spearman ρ
0.3
0.2
0.1
0.0
−0.1
−5 −4 −3 −2 −1 0 1 2 3 4 5
k (days): sentiment of day d−k vs return of day d
k<0: price leads sentiment · k>0: sentiment leads price



7. Live forecast (from the latest close)

Horizon	Date	Forecast	80% low	80% high	Random walk
n+0	2026-11-04	$77,261	$75,313	$79,259	$77,345
n+1	2026-11-05	$77,220	$74,406	$80,140	$77,345
n+2	2026-11-06	$77,303	$73,690	$81,093	$77,345
n+3	2026-11-07	$77,596	$73,222	$82,231	$77,345
n+4	2026-11-08	$77,630	$72,480	$83,146	$77,345
n+5	2026-11-09	$77,764	$71,951	$84,047	$77,345

8. Limitations

- Only 117 out-of-sample trading days: an annualised Sharpe ratio estimated on that sample has a standard error of roughly ±1.8 (≈√(365/N)), so a Sharpe ratio within about ±3.5 of zero (or of buy & hold) is not statistically distinguishable from it.
- Each day's sentiment is a capped sample of posts (top or latest), not the full X firehose; post-count features therefore measure sampling more than true volume.
- Lexicon sentiment misses sarcasm and context; FinBERT and CryptoBERT are available via `SENTIMENT_ENGINE=finbert` or `cryptobert` for comparison.
- Results depend on one asset and one period; no hyper-parameters were tuned on the test window.

9. Reproduce

```
pip install -r requirements.txt
cp .env.example .env   # add APIFY_TOKEN and TAVILY_API_KEY
python run_pipeline.py all
```